

Exam 1

CH368 – Biophysics

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This is provided as a MS Word file but please turn in your exam responses as a PDF on Canvas. Feel free to copy/paste the questions onto a new file or if you prefer to write your answers by hand, scan the paper and convert the images to a PDF.

Show all your work and write all your explanations in full sentences.

Honor Code: The core values of The University of Texas at Austin are learning, discovery, freedom, leadership, individual opportunity, and responsibility. Each member of the university is expected to uphold these values through integrity, honesty, trust, fairness, and respect toward peers and community.

Total Points:

/ 150

1. (15 points) **Boltzmann Game.** In the “Boltzmann Game” simulation, you were asked to compute the entropy of the system as function of the number of rounds. In a separate question, you were asked to explore the effect of preventing players with a certain number of coins to receive additional ones. Imagine you run a simulation with each player starts with 5 coins, and you set the cap at 20 coins (i.e. players with >20 coins cannot get more). Do you expect this to have an effect on the entropy of the system? Explain your thought process.

I would expect the maximum entropy of the system to decrease. If we cap the number of coins that each player can have, there is a smaller number of microstates that the system can access. When uncapped, there is a probability that some players can hoard many coins on their own. Although unlikely, accessibility to these low probability microstates increases the maximum entropy of the system.

2. (15 points) **Biophysics:** “What does the earth get from the sun?” Explain in your own words.

Earth gains low-entropy solar radiation from the Sun and emits high entropy infrared radiation back into space. This entropy gradient provides the free energy that allows for systems on Earth to do work.

3. (15 points) **Gas diffusion.** Consider this scenario: Your friend makes the following claim: “In an ideal gas, if you double the volume of the container at constant temperature, the entropy doubles”. Your friend argues that since entropy is an extensive quantity, this is equivalent to bringing two containers of the same gas together, and simply opening a valve between them, so the entropy should be twice the initial entropy. Is your friend correct or not? Please use equations in your argument. What is the entropy difference when the gas expands to twice its volume at constant temperature?

The friend is wrong in this case. The total amount of entropy of the system would increase, but it would not double.

Given that each particle is independent, the total number of microstates is the product of the choices for which each particle can reside. Thus, the number of microstates increases by 2^N .

By the Boltzmann Equation,

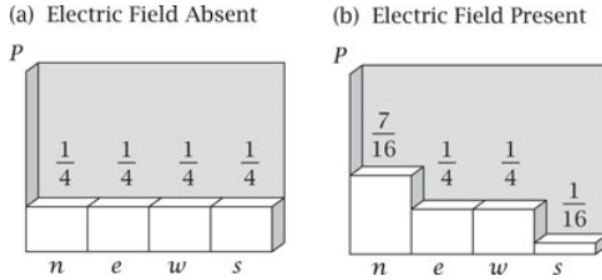
$$S = k_B \ln(W)$$

Where W is the number of microstates in the system. By this definition, for a gas expanding from $V \rightarrow 2V$ at constant T , the change in entropy is

$$\Delta S = k_B \ln(2^N) = Nk_B \ln(2)$$

4. (30 points) **Entropy:** Consider an electric dipole in a 2D plane (perhaps think of this as a “compass”). The dipole can point in any of four directions (north, east, west, south). In the

absence of an electric field, the dipole can be found pointing in each direction an equal portion of the time (a). However, when the electric field is turned on, the dipole points mostly north but may be found pointing in other directions a certain portion of the time (b). Calculate the interaction energy (ie. enthalpy) between the dipole and the electric field given that the system is at room temperature? Explain your reasoning.



To assign energies to each of the four orientations of the dipole, we can consider:

$$U = -\vec{p} \cdot \vec{E}$$

In this 2D model, the north is aligned with the electric field, the south is anti-aligned, and east and west are perpendicular to the field. We can use the Boltzmann distribution because this is a system at thermal equilibrium where orientation probabilities are set by a competition between energy and entropy.

Given the Boltzmann distribution,

$$P_i = \frac{1}{Q} e^{-\beta \epsilon_i}, \quad \beta = \frac{1}{k_B T}, \quad Q = \sum_{i=1}^t e^{-\beta \epsilon_i}$$

Let,

$$Q = e^{-\beta U_{north}} + e^{-\beta U_{south}} + e^{-\beta U_{east}} + e^{-\beta U_{west}}$$

Given the energies of each orientation,

$$U_{north} = -pE$$

$$U_{south} = +pE$$

$$U_{east} = 0$$

$$U_{west} = 0$$

Inserting these energies into the sum gives,

$$Q = e^{\beta pE} + 2 + e^{-\beta pE}$$

Because $U_{north} = -pE$, the Boltzmann factor becomes $e^{+\beta pE}$.

Plugging in Q and using P_{north} ,

$$\frac{7}{16} = \frac{e^{\beta pE}}{e^{\beta pE} + 2 + e^{-\beta pE}}$$

Solving at room temperature ($T = 298\text{K}$)

$$pE = 4.0 \times 10^{-21} \text{ J}$$

The mean interaction energy (enthalpy) is

$$\langle U \rangle = \sum_{i=1}^t P_i U_i$$

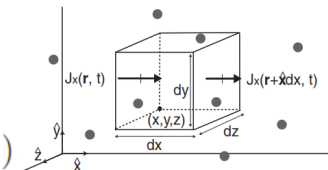
Numerically,

$$\langle U \rangle = -1.5 \times 10^{-21} \text{ J}$$

5. (15 points) Diffusion. You would like to measure the diffusion rate of a certain biomolecule in water so you can estimate its hydrodynamic radius. You design an experiment that uses a fluorescent label attached to your biomolecule, and you plan to track the 2D diffusion of this molecule using a fluorescence microscope. Unfortunately, the time resolution of your microscope is too slow, so you only see the molecule for 1-2 frames before it diffuses out of the focus. Aside from purchasing a faster camera for your microscope, explain how you would modify your experiment to be able to measure the hydrodynamic radius of this molecule.

To decrease the speed at which the molecule diffuses out of frame, we can reduce the diffusion coefficient, D , which allows the molecule to be tracked for more frames. This can be achieved by either lowering the temperature or increasing the viscosity of the solution but lowering the temperature would be the simplest approach. Slower diffusion allows more accurate measurement of the diffusion, which can then be used with the Stokes-Einstein equation to estimate the hydrodynamic radius of the molecule.

6. (10 points) Fickian Diffusion: In the diffusion lecture we noted that in Fick's first law the Flux (J) is proportional to the first derivative of the density, as shown below. However, in the computer code we used the second derivative of the density to compute the time evolution (see below, line 6 in the code snippet). Was there a mistake in the code? If not, please explain why the equations in the code are different from the equations we used in lecture.

$$J_u(\mathbf{r}, t) = -D \frac{\partial \rho(\mathbf{r}, t)}{\partial u} \quad (u = x, y, z)$$


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1 # Propagate the distribution in time
2 for time in range(numTimeSteps):
3     # Assign a variable to the current distribution
4     currentDistribution = distribution[:, time]
5     # Calculate the second derivative
6     secondDeriv[:, time] = np.gradient(np.gradient(currentDistribution, xPosition), xPosition)
7     # Convolve with a smoothing filter can help reduce noise
8     secondDeriv[:, time] = np.convolve(secondDeriv[:, time], np.ones(20) / 20, mode='same')
9
10    # Assign a variable to the current second derivative
11    currentSecondDeriv = secondDeriv[:, time]
12
13    # Move particles based on the second derivative and diffusion coefficient
14    for idx in range(len(xPosition) - 1):
15        # Check if the current particle and its neighboring particle have enough density to diffuse
16        if (currentDistribution[idx] > abs(diffusionCoeff * currentSecondDeriv[idx])) and (currentDistribution[idx + 1] > abs(diffusionCoeff * currentSecondDeriv[idx])):
17            # Move particles from the current position to the neighboring position
18            currentDistribution[idx + 1] = currentDistribution[idx + 1] + (diffusionCoeff * currentSecondDeriv[idx])
19            currentDistribution[idx] = currentDistribution[idx] - (diffusionCoeff * currentSecondDeriv[idx])
20

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There is no mistake in the code. Fick's first law relates the particle flux to the spatial gradient of the density, while Fick's second law describes how the density evolves in time. These two equations address different physical questions. The time evolution of the density is obtained using Fick's first law with mass conservation in absence of any chemical reactions, which leads to a second spatial derivative of the density. In the code, the goal is to compute how the density distribution changes over time, so the second derivative appears naturally through Fick's second law and represents the effect of curvature in the density on particle motion.

7. (15 points) Ideal gas equation. Most students learn the ideal gas equation as an “empirical” equation that was postulated based on observations. However, in lecture we were able to derive the ideal gas equation based on a simple model and taking advantage of the fundamental thermodynamic relations (i.e. Maxwell's equations).

a. Explain, in your own words, what are Maxwell's equations and where they come from.

Maxwell's relations are a way to relate different quantifiable thermodynamic properties of a system to one another through partial derivatives taken with constant variables. These arise from fundamental thermodynamic relations by expressing state functions such as internal energy in terms of their natural variables. This allows hard-to-measure quantities such as entropy to related experimentally to variables which can easily be measured.

b. Explain what assumptions we made in deriving the ideal gas equation.

In deriving the ideal gas equation, we assumed that gas particles occupy a lattice with M available sites, where each site can be occupied by at most one of the N particles. We further assumed that the particles do not interact with one another, except through elastic collisions, and that the gas is in thermal equilibrium. To simplify the statistical description, we used Stirling's approximation for factorials and a Taylor series expansion of $\ln(1 - x)$, which is valid in the dilute limit $N \ll M$.

8. (15 points) Diffusion. Does the following statement agree or disagree with what we learned in class? (explain briefly) Diffusion causes molecules to move from regions of high concentration to regions of low concentration because the high number of collisions between molecules in the region of high concentration “pushes” the molecules towards the regions of lower concentrations. This is similar to how billiard balls become randomly dispersed across a pool table on the first turn when the player “breaks” the initial triangular configuration of the balls.

This is not quite consistent with what we learned in class. Brownian motion is the random movement of particles in a given region. The diffusion of particles from high to low concentrations is not caused by the collisions “pushing” these particles to low concentration areas but is purely the emergent result of the probability of net movement from high to low concentration arising from random motion of many particles.

9. (20 points) Diffusion. Assume that the amount of time it takes for a grain of sugar to dissolve in a cup of coffee is directly proportional to the time it takes for the sugar molecules to diffuse away from the area immediately near the sugar crystal into the rest of the volume of the liquid. If you put a grain of sugar in a cup of iced coffee at 10 °C and it takes 30 min for it to dissolve (without stirring), how long do you estimate it will take if you dissolve the same grain of sugar in hot coffee at 65 °C? What relationship do you use to make this estimate?

Using the Stokes-Einstein Relation

$$D = \frac{k_B T}{6\pi\eta r},$$

The diffusion of the sugar will increase due to the increasing temperature. The viscosity of water also decreases as the temperature increases. The viscosity of water at 10° and 65° were given by Kestin *et. al.* (*J. Phys Chem Ref Data*, 2009). The ratio of the diffusion coefficient is shown by,

$$\frac{D_{10^{\circ}C}}{D_{65^{\circ}C}} = \frac{283 \left(\frac{1}{\eta_{10^{\circ}C}}\right)}{338 \left(\frac{1}{\eta_{65^{\circ}C}}\right)} = 0.27$$

Since,

$$t \propto \frac{1}{D}$$

The sugar in the 65°C cup of coffee will have a dissolution time that is approximately 3.7 times shorter than in the 10 °C cup. Thus, it would take around 8 minutes for it to dissolve fully in the hotter cup.